Pengenalan Pembelajaran Mesin

RTI235004 – Pembelajaran Mesin

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- Jika terdapat merek ataupun brand dalam materi ini, merek dan brand tersebut merupakan miliki pemilik merek dan brand.

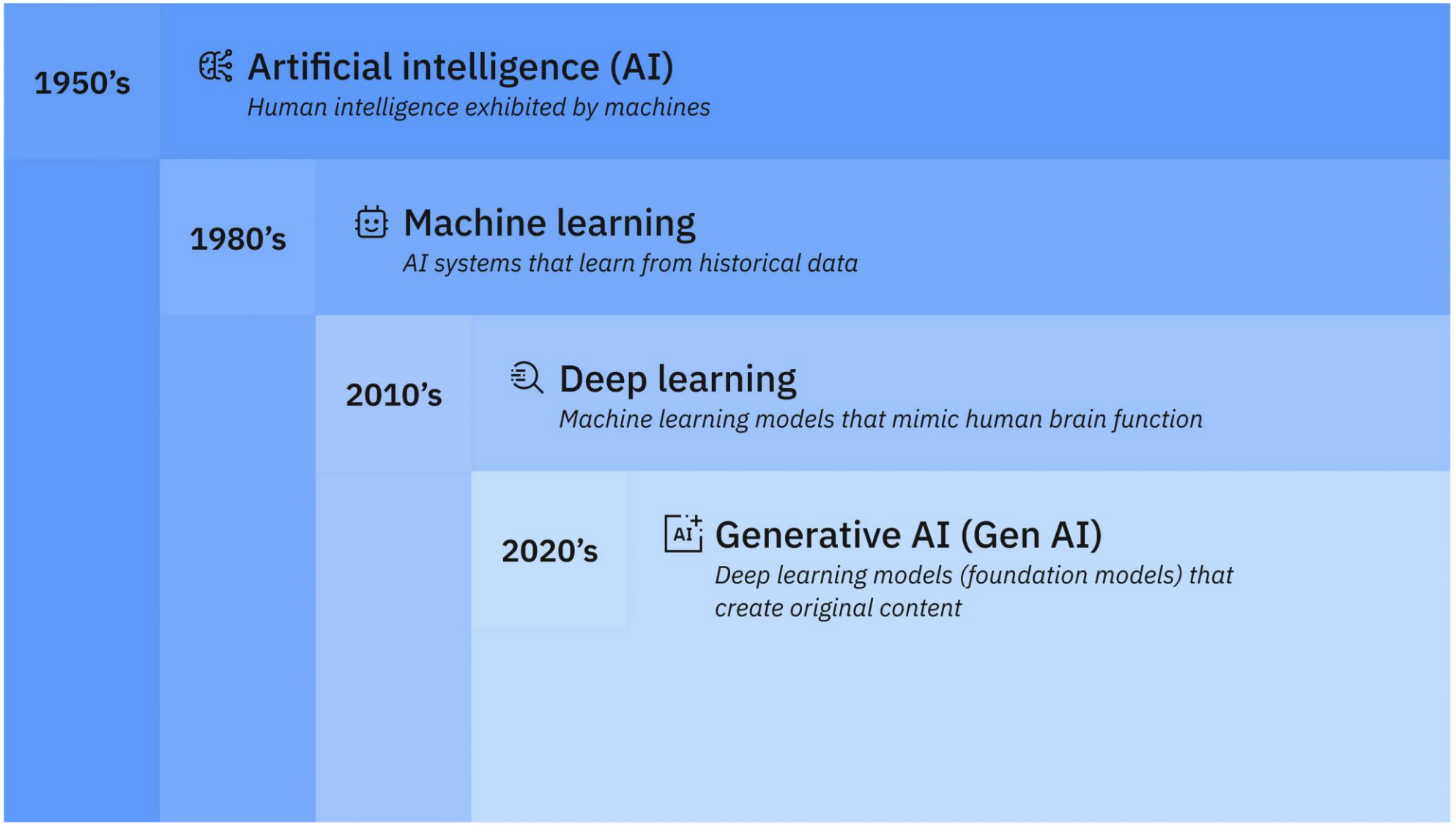
Outlines

- Apa itu pembelajaran mesin?
- Tipe Pembelajaran Mesin
- Siklus Pengembangan Model Pembelajaran Mesin
- Etika dan Tantangan

Apa itu pembelajaran mesin?

Mengapa saat ini sangat populer?

AI vs. ML vs. DL vs. GenAI



Lalu, apa itu pembelajaran mesin?

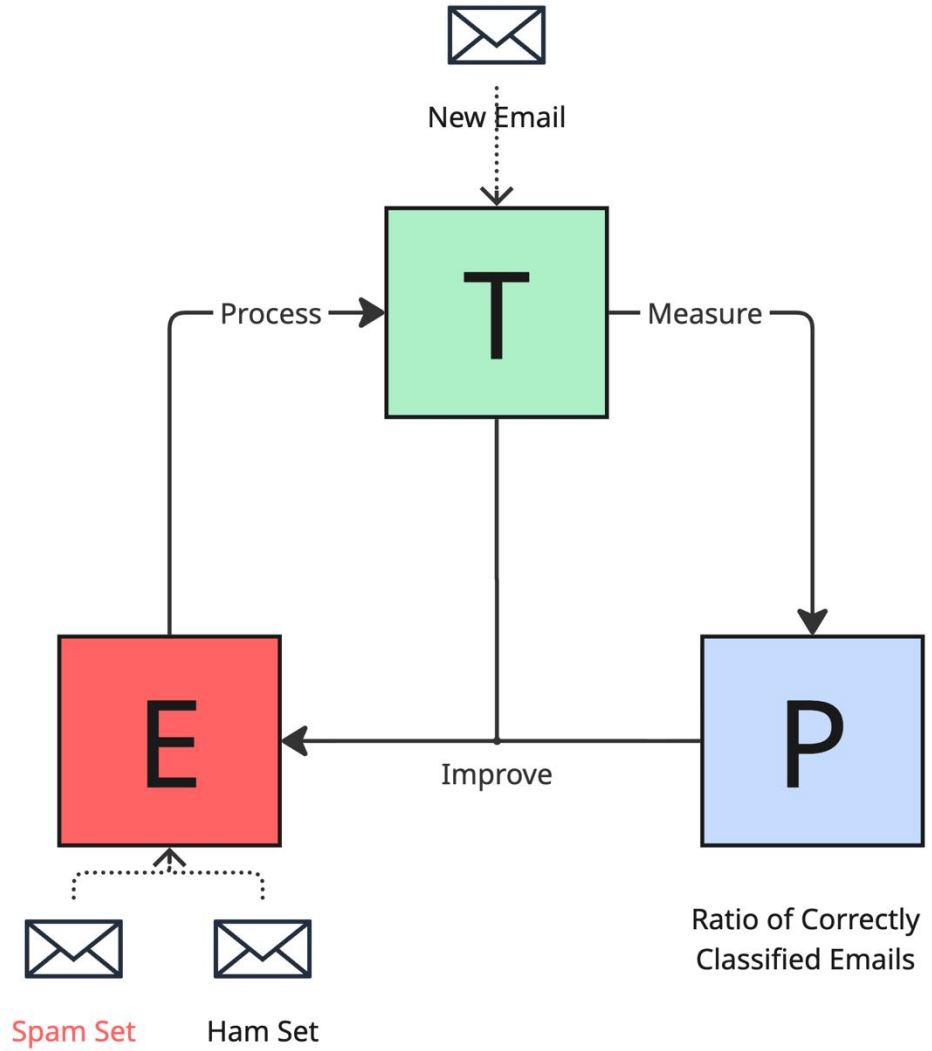
Machine learning is the science (and art) of programming computers so they can learn form data – Aurelien Geron, 2019

A field of study that gives computers the ability to learn without being explicitly programmed – Arthur Samuel, 1959

A computer program is said to learn from experience (E) with respect to some task (T) and some performance measure (P), if its performance on T, as measured by P, improved with experience E – Tom Mitchell, 1997

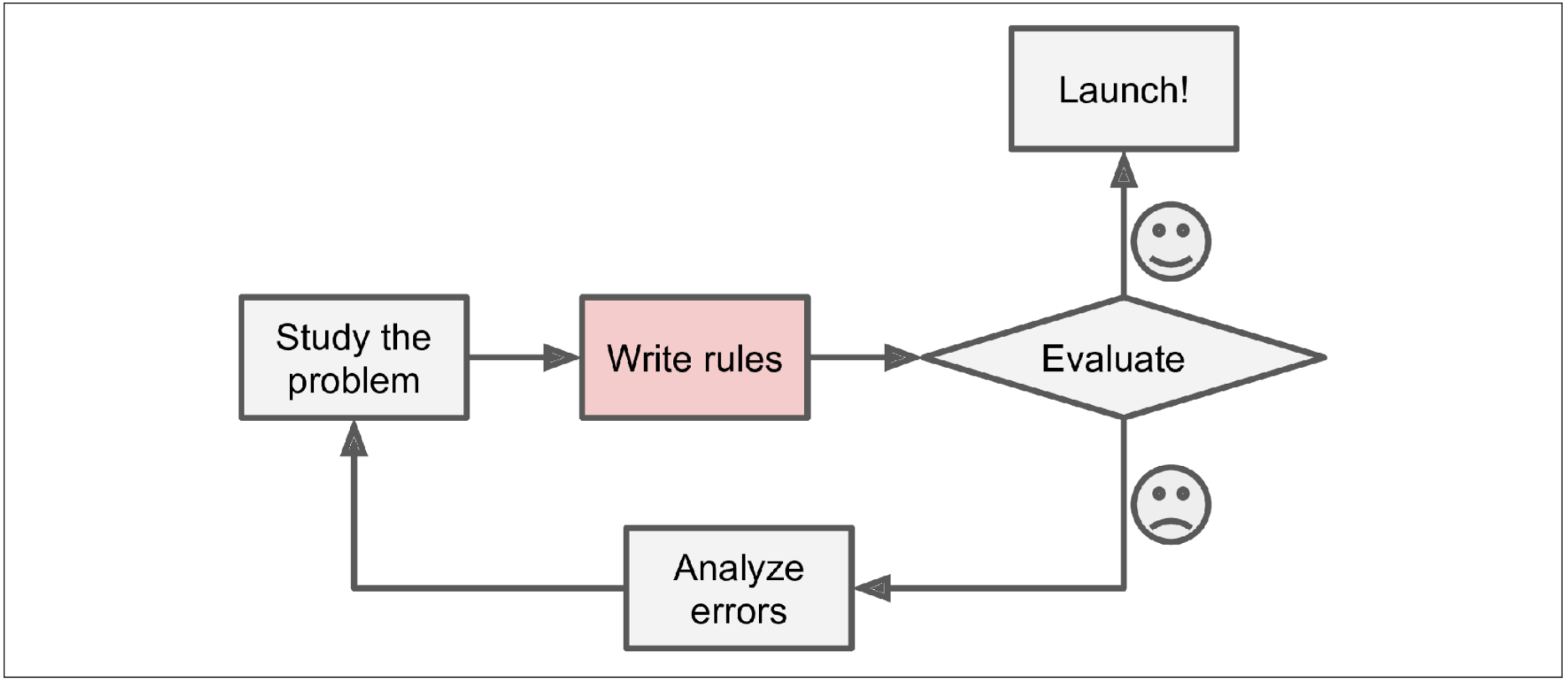
Konsep pembelajaran mesin dari Tom Mitchell

Kasus Filter Email



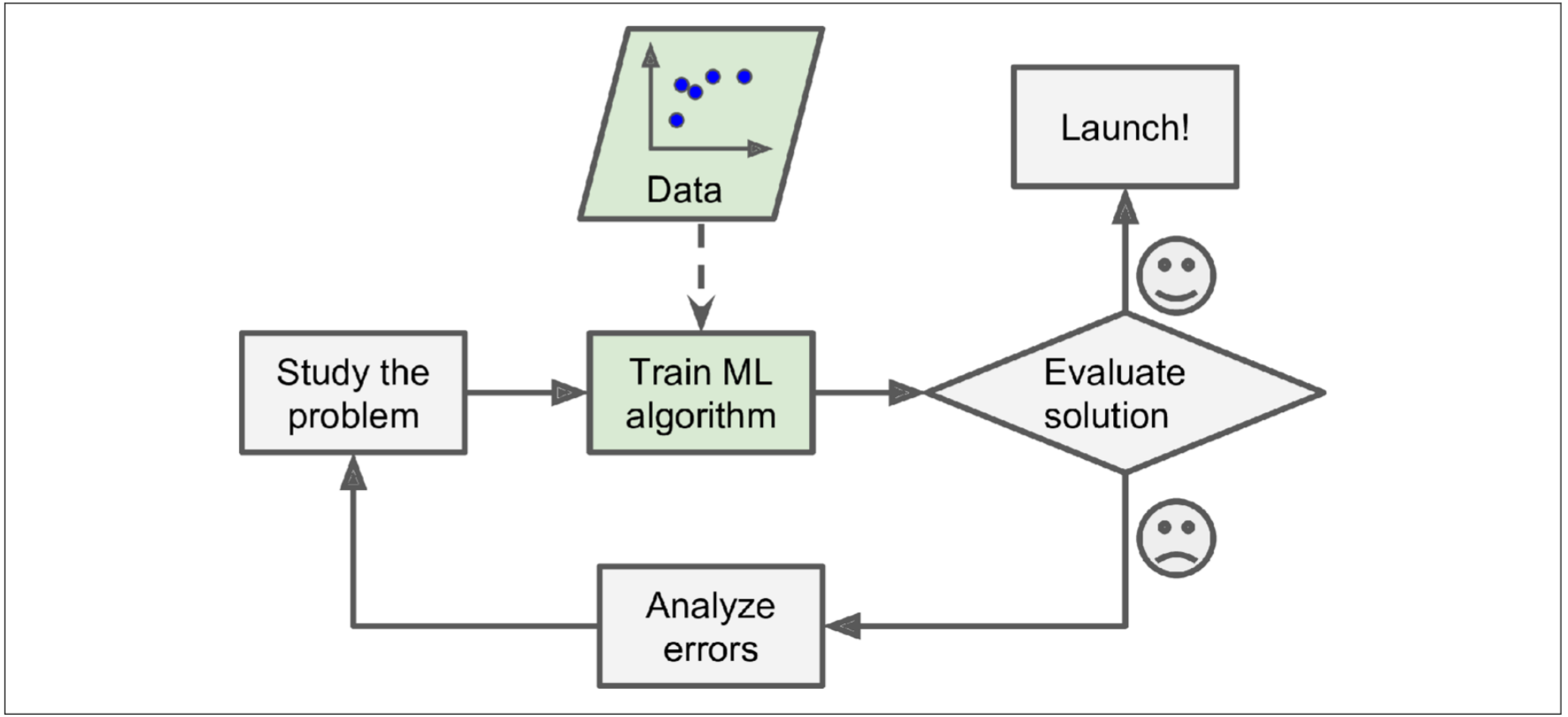
Mengapa menggunakan pembelajaran mesin? #1

Kasus Filter Email – Pendekatan Tradisional



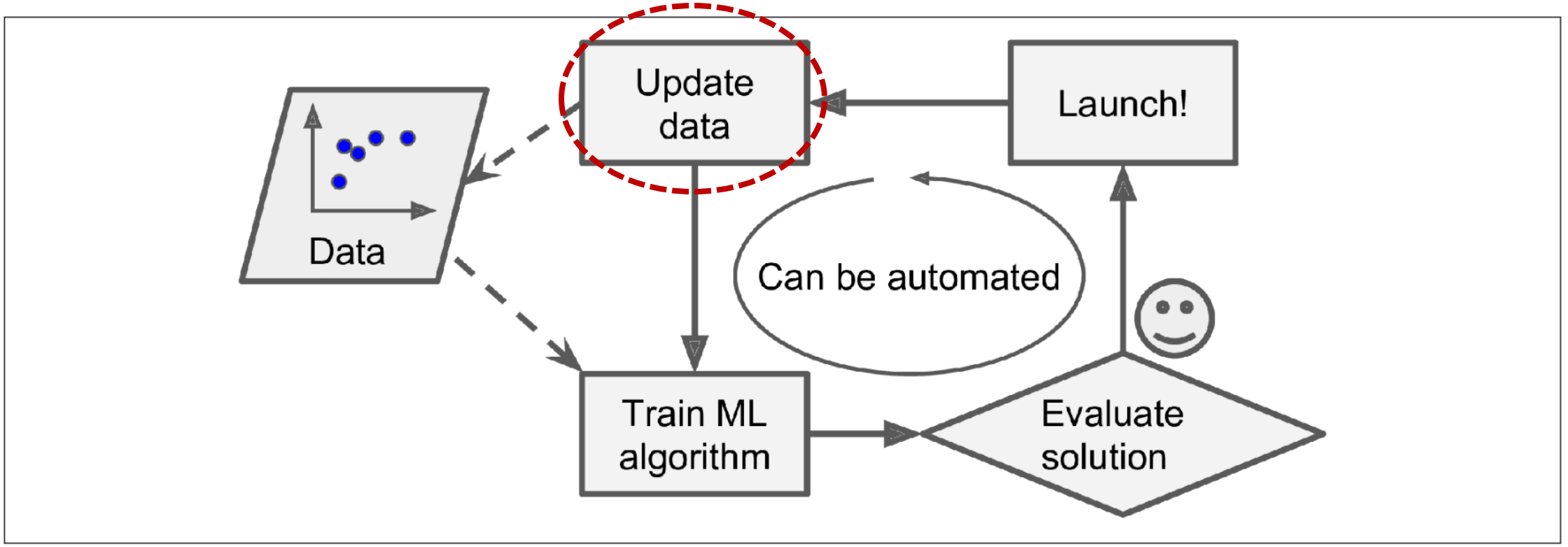
Mengapa menggunakan pembelajaran mesin? #2

Kasus Filter Email – Pendekatan Pembelajaran Mesin (Konsep)



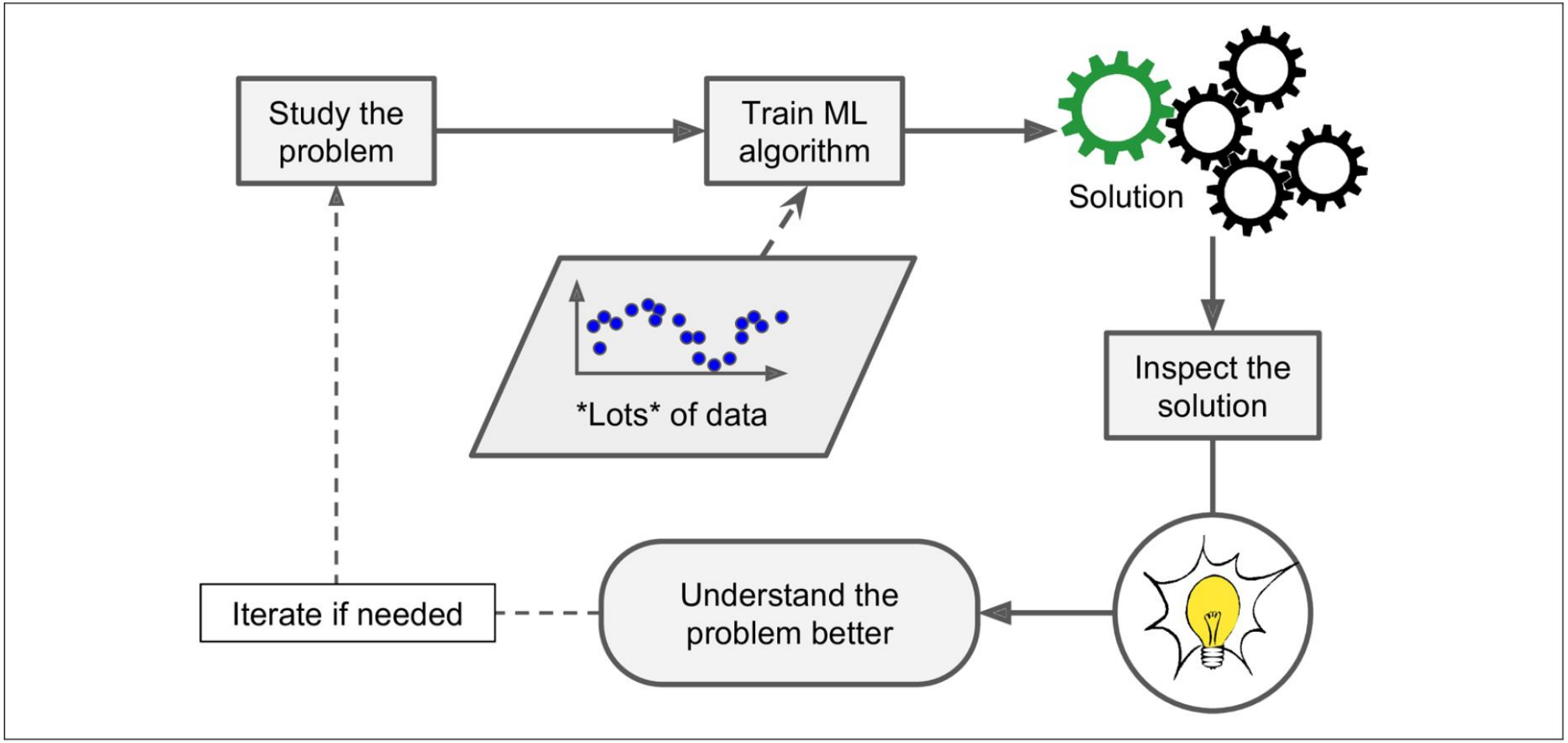
Mengapa menggunakan pembelajaran mesin? #3

Kasus Filter Email – Pendekatan Pembelajaran Mesin → Pola Berubah?



Mengapa menggunakan pembelajaran mesin? #4

Kasus Filter Email – Pemahaman Lebih Lanjut? Data Besar? → Data Mining



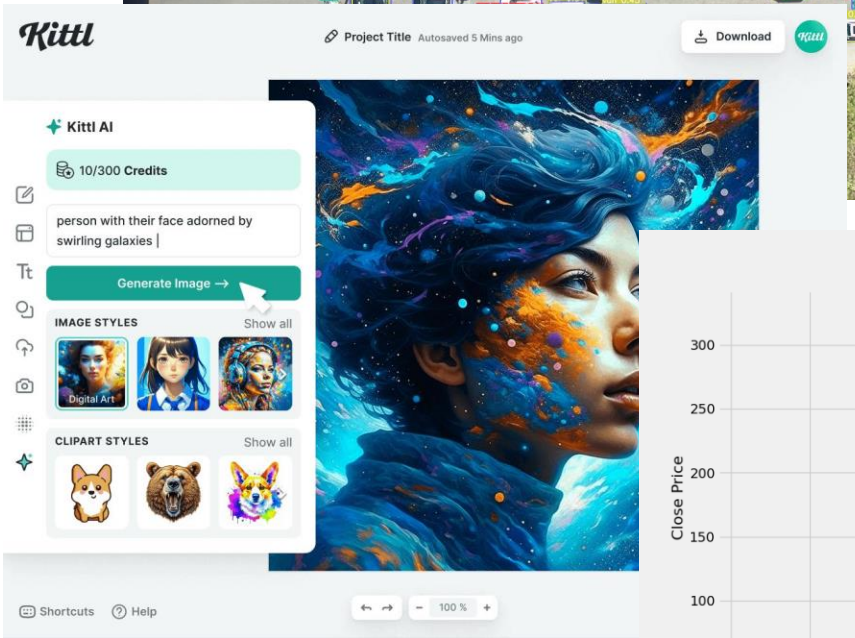
Mengapa menggunakan pembelajaran mesin? #5

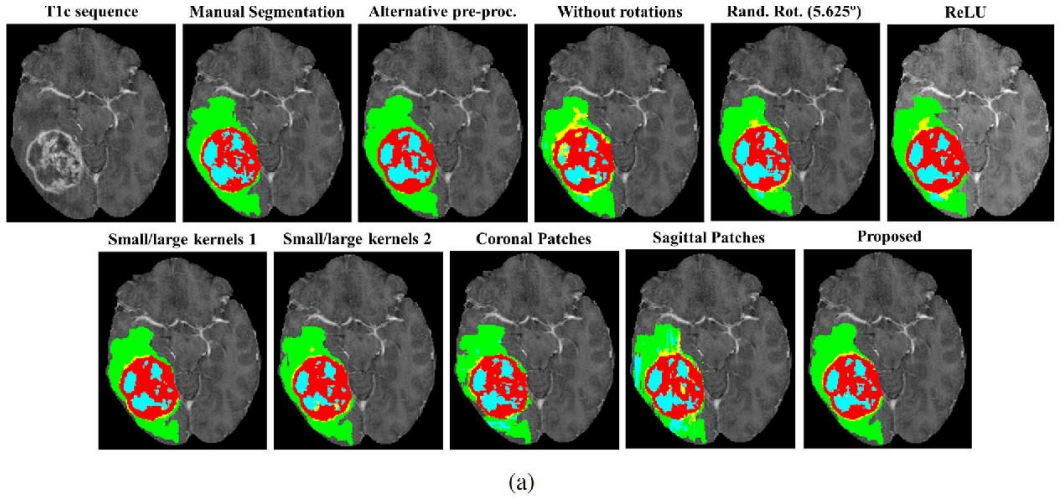
Rangkuman

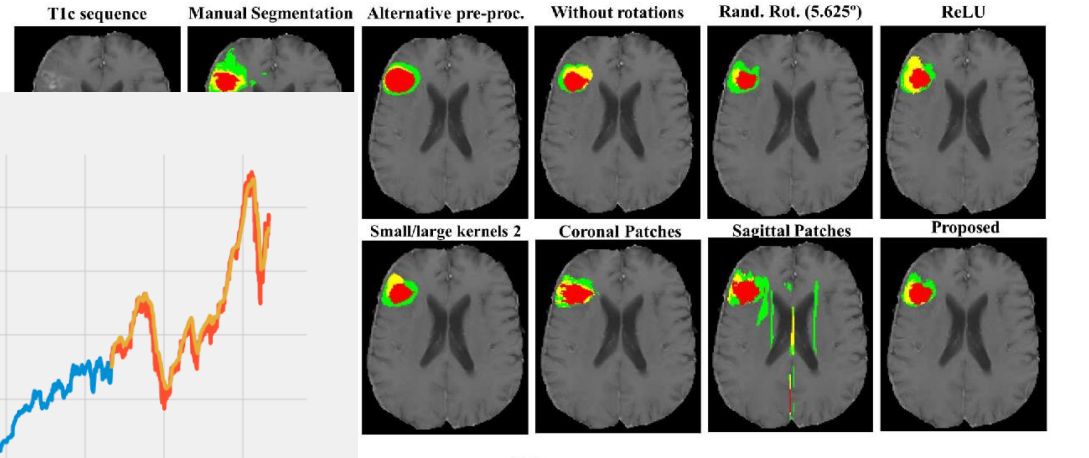
- Permasalahan yang akan diselesaikan memerlukan banyak aturan (rules) untuk diselesaikan.
- Permasalahan yang kompleks yang sulit diselesaikan dengan pendekatan tradisional.
- Perubahan lingkungan (data) yang fluktuatif.
- Mendapatkan “insight” dari permasalahan yang kompleks dengan data yang besar.

Contoh pemanfaatan pembelajaran mesin

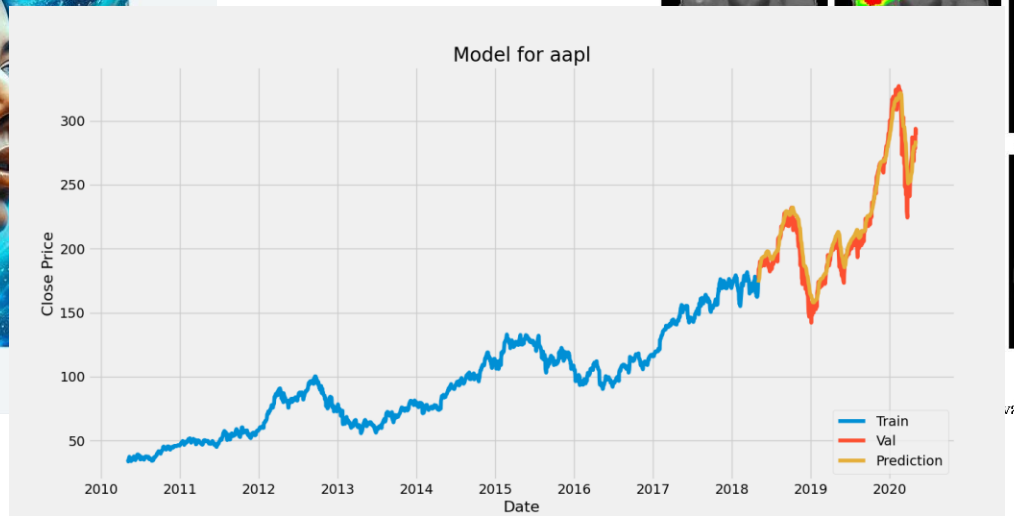






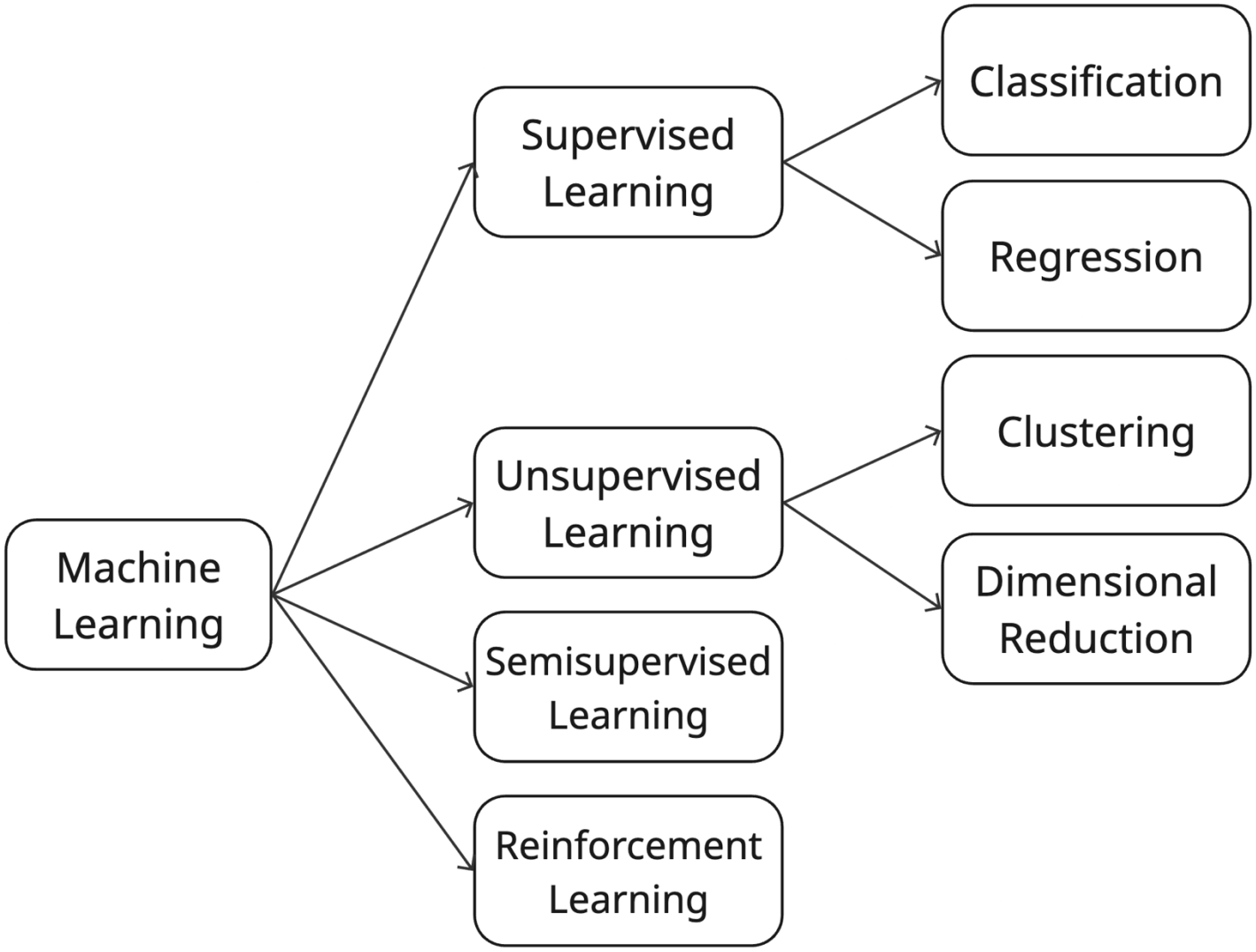


validation, showing the effect of each component of the proposed method. In the first row, we have a



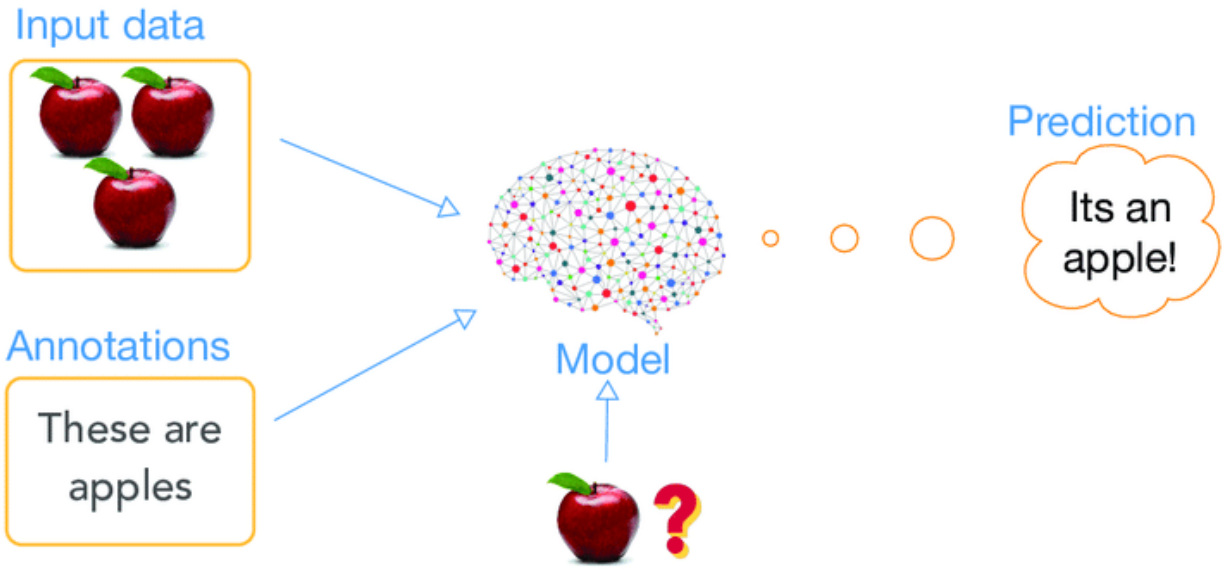
Tipe Pembelajaran Mesin

Tipe Pembelajaran Mesin

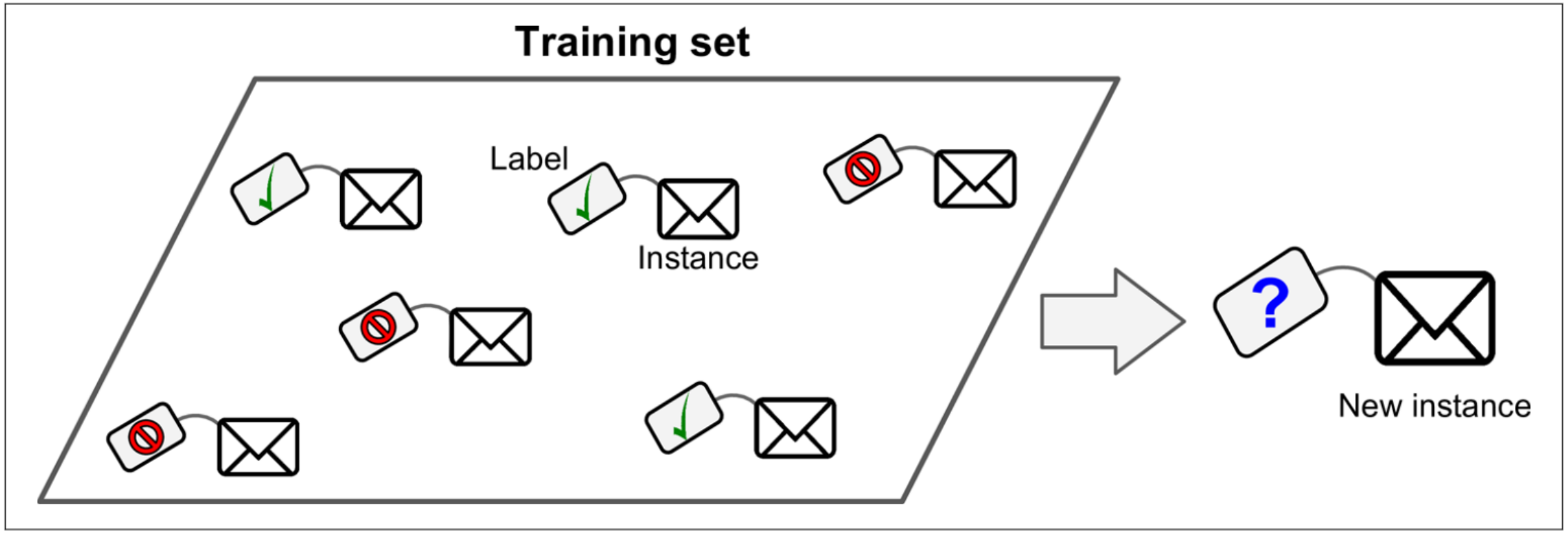


Supervised learning

- Membutuhkan data latih beserta dengan labelnya sebagai media pembelajaran algoritma.
- Digunakan untuk tugas
  - Klasifikasi → prediksi label
  - Regresi → prediksi nilai



Supervised learning - Klasifikasi

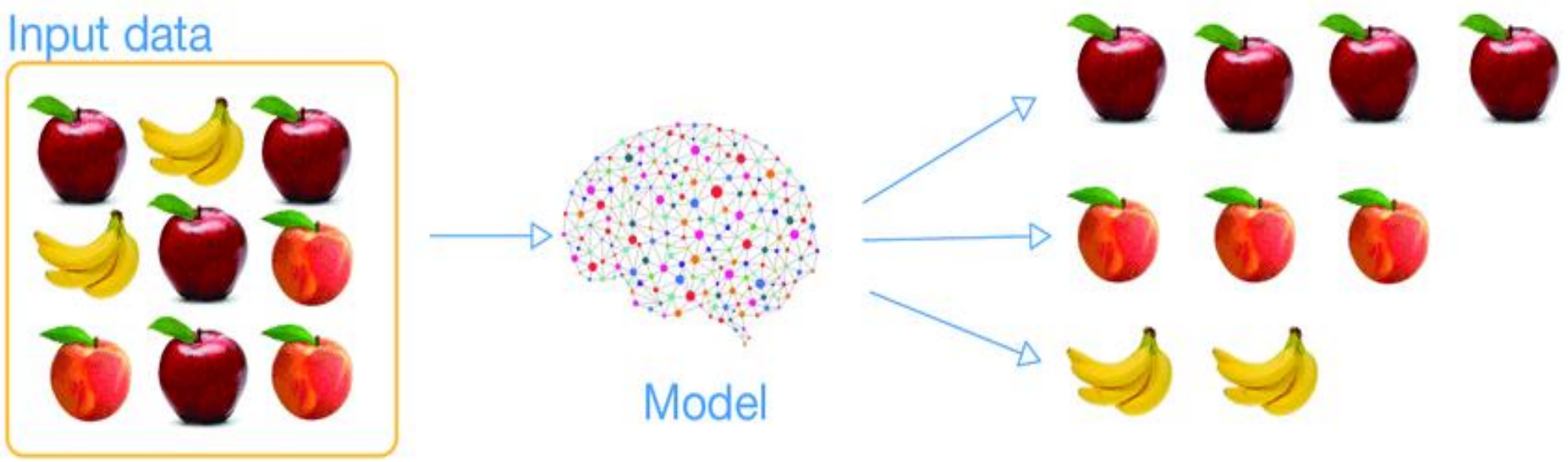


Supervised learning - Regresi

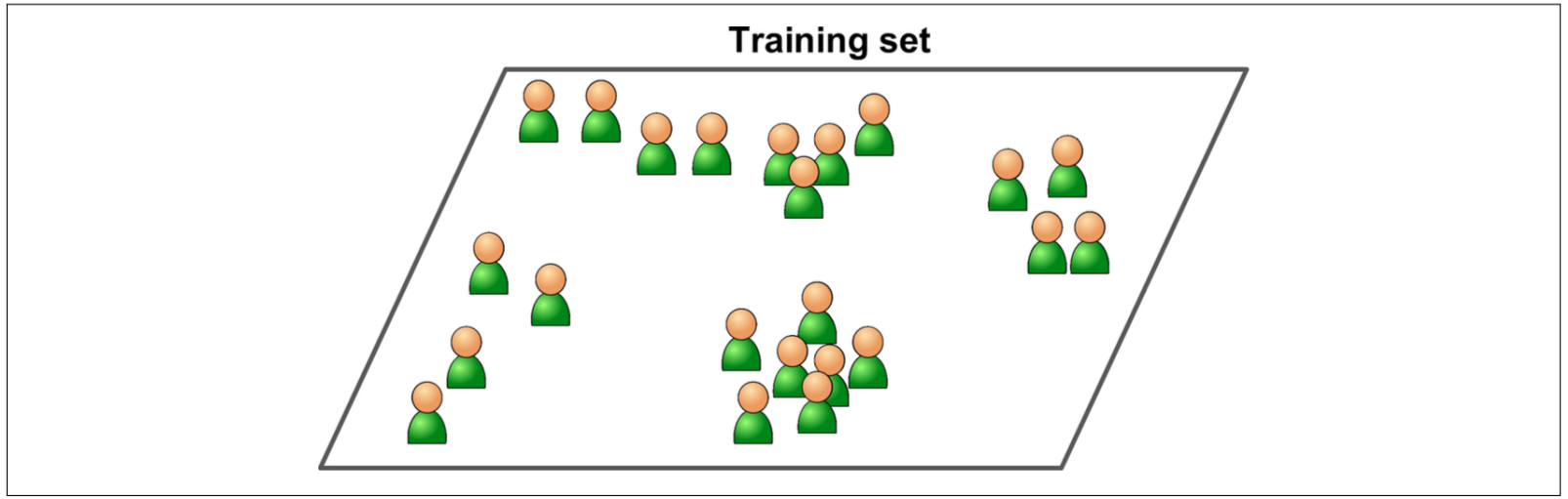


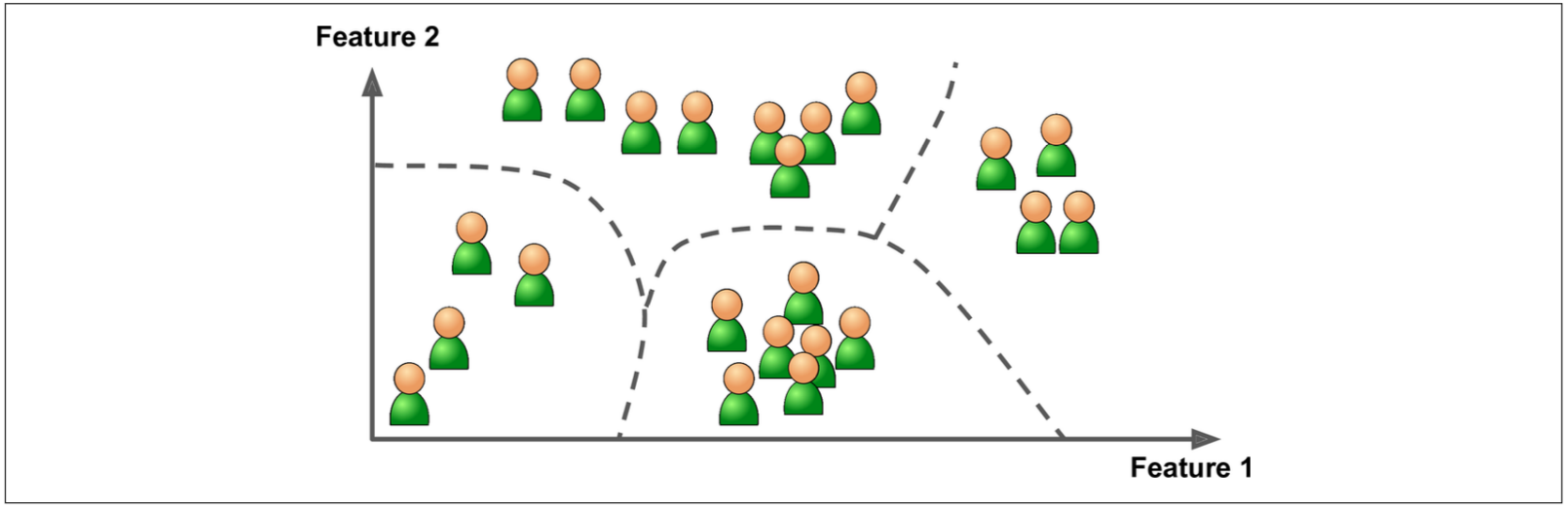
Unsupervised learning

- Algoritma melakukan pembelajaran tanpa memerkulan label.
- Data akan **dikelompokkan** berdasarkan **kesamaan karakteristik**.

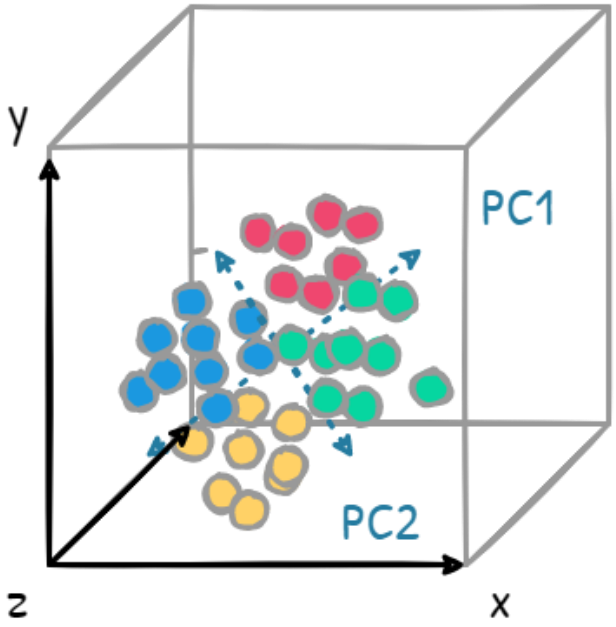


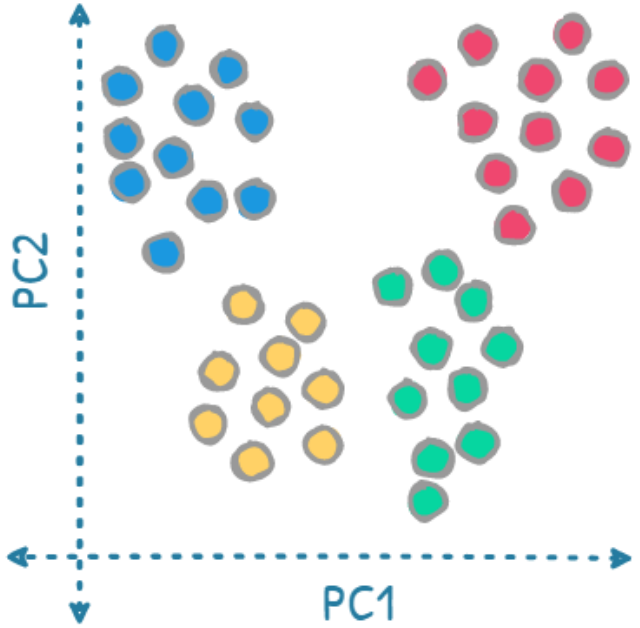
Unsupervised learning - Klasterisasi

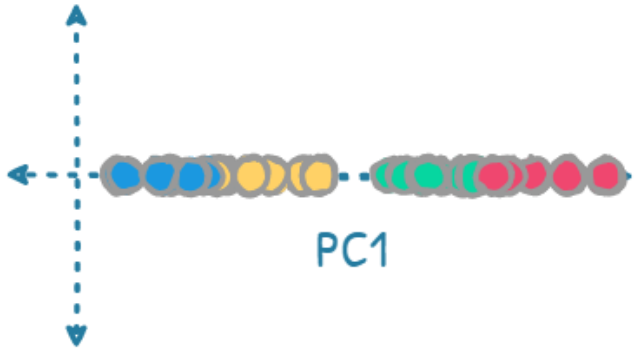




Unsupervised learning – Dimensional Reduction

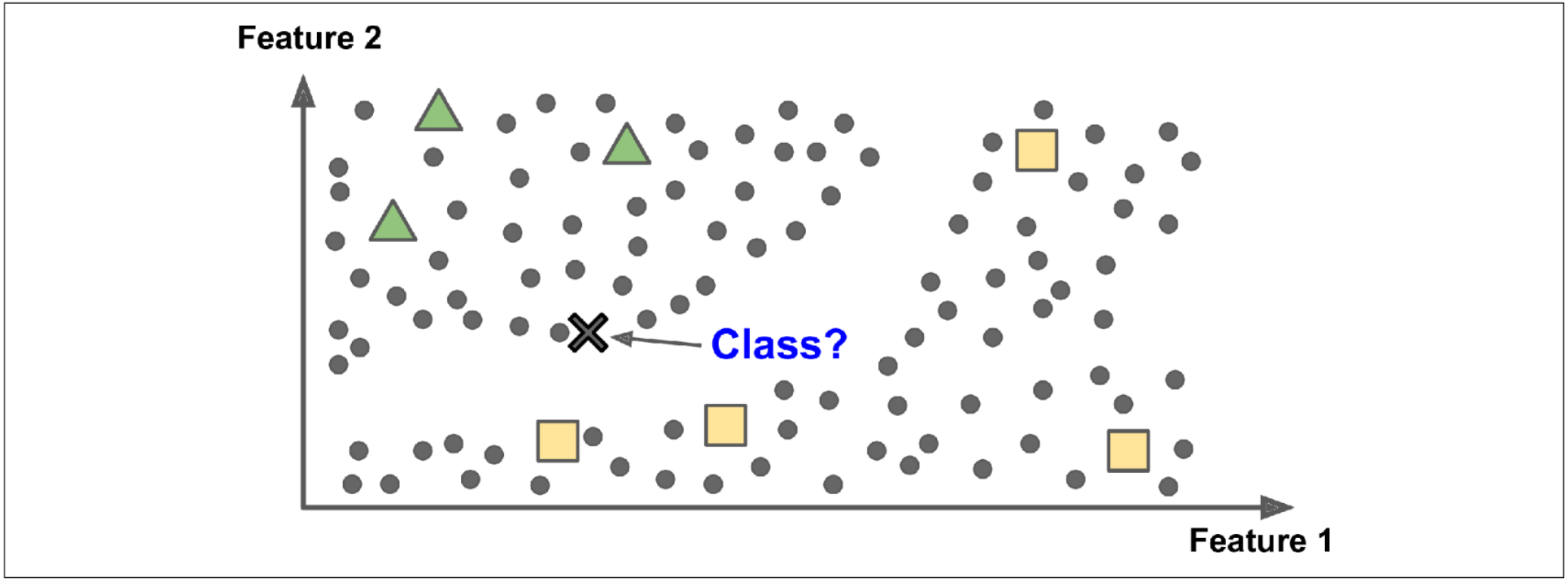






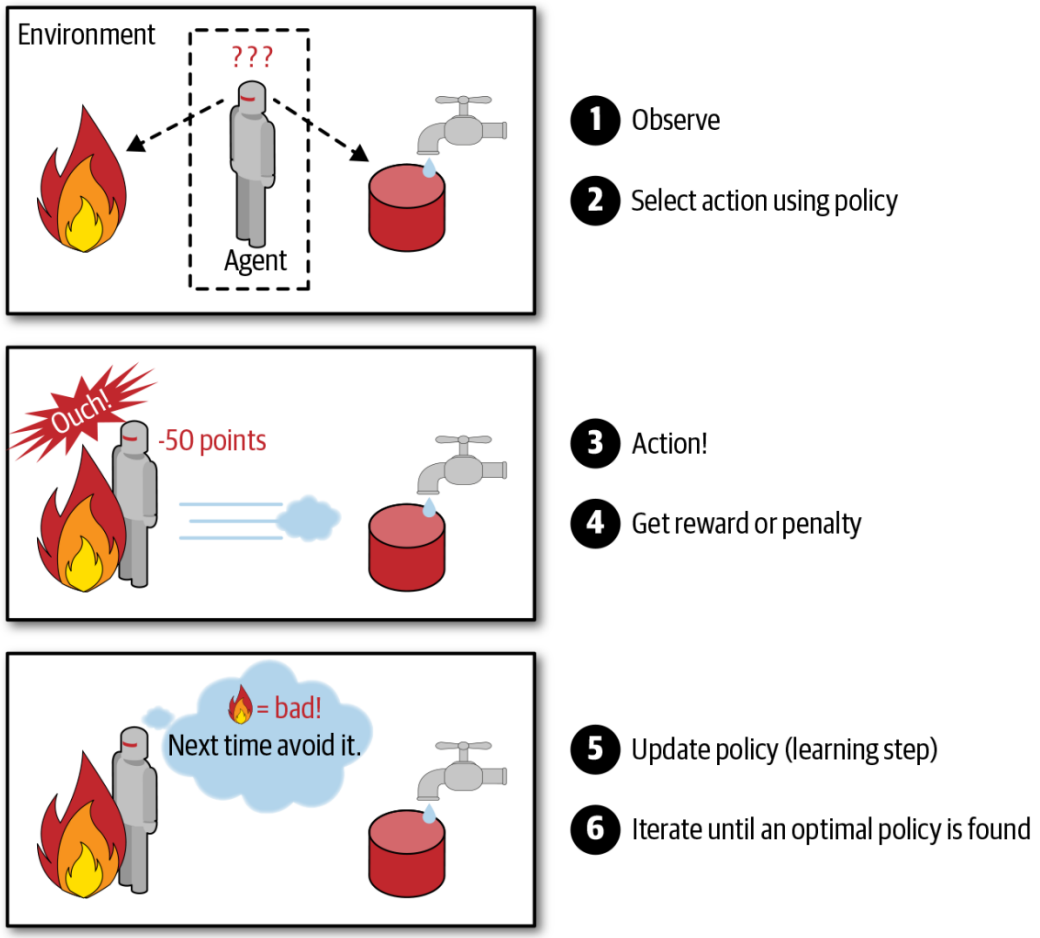
Semisupervised learning

- Gabungan antara *supervised* dan *unsupervised learning* untuk mendapatkan keunggulan dari keduanya.
- Digunakan dalam kasus data banyak, hanya sedikit yang terlabeli

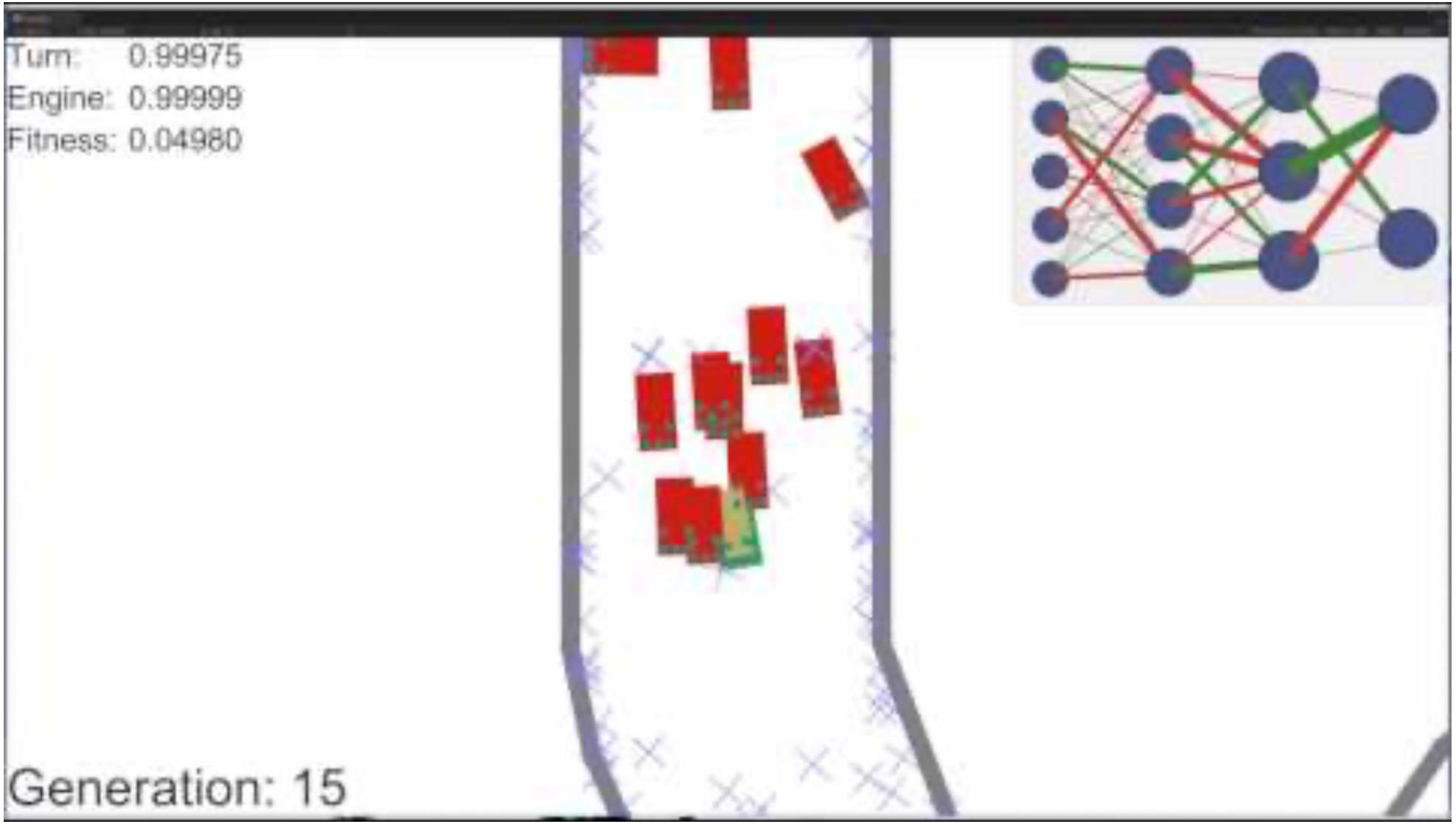


Reinforcement learning

- Menggunakan "**agen**" sebagai aktor utama dan melakukan pembelajaran **secara langsung terhadap lingkungan**.
- Dilakukan secara berulang untuk mendapatkan hasil yang diharapkan.
- Menggunakan konsep "***reward***" dan "***punishment***"
- Reward → Jika sesuai
- Punishment → Jika tidak sesuai



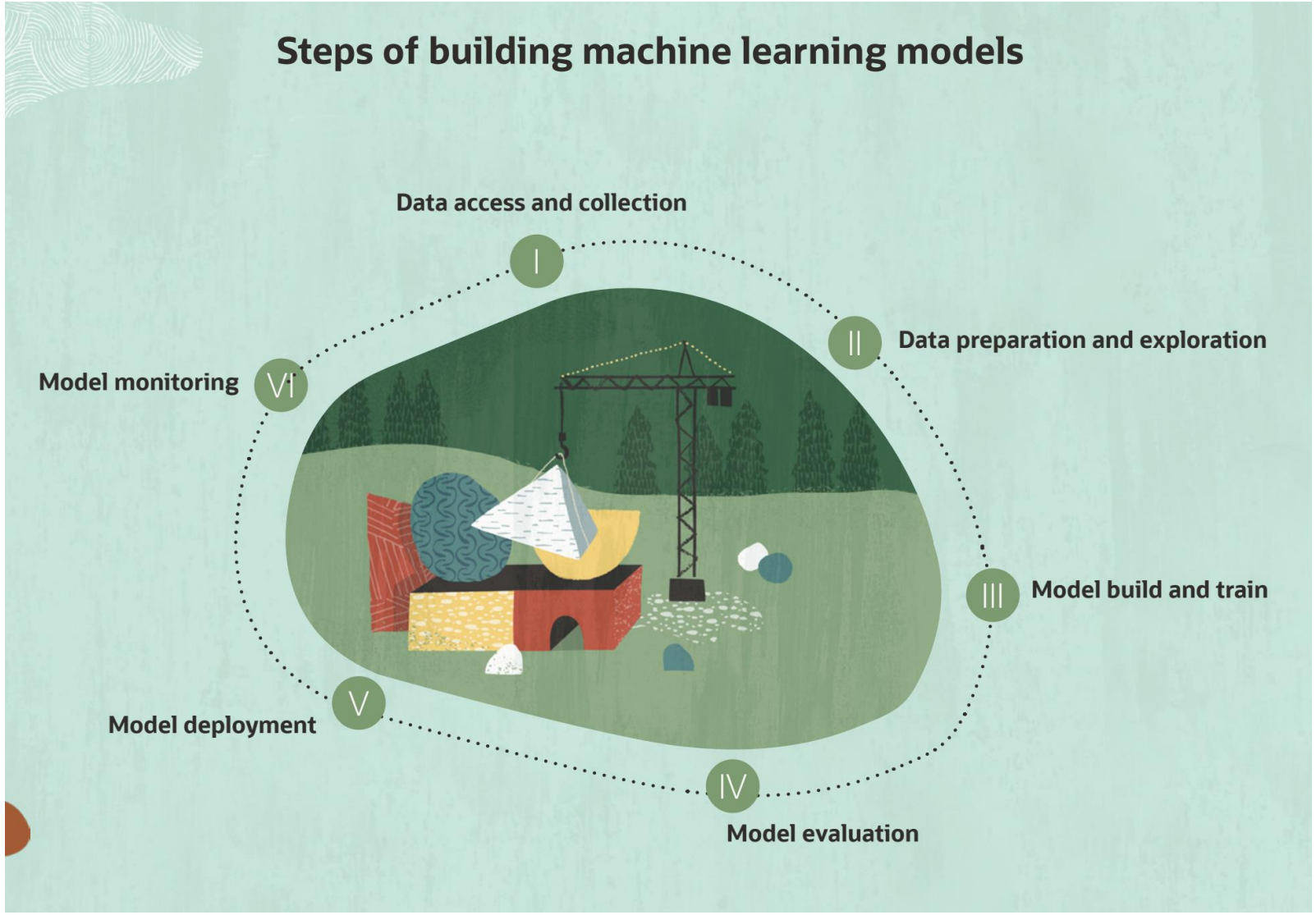
Reinforcement learning – Contoh



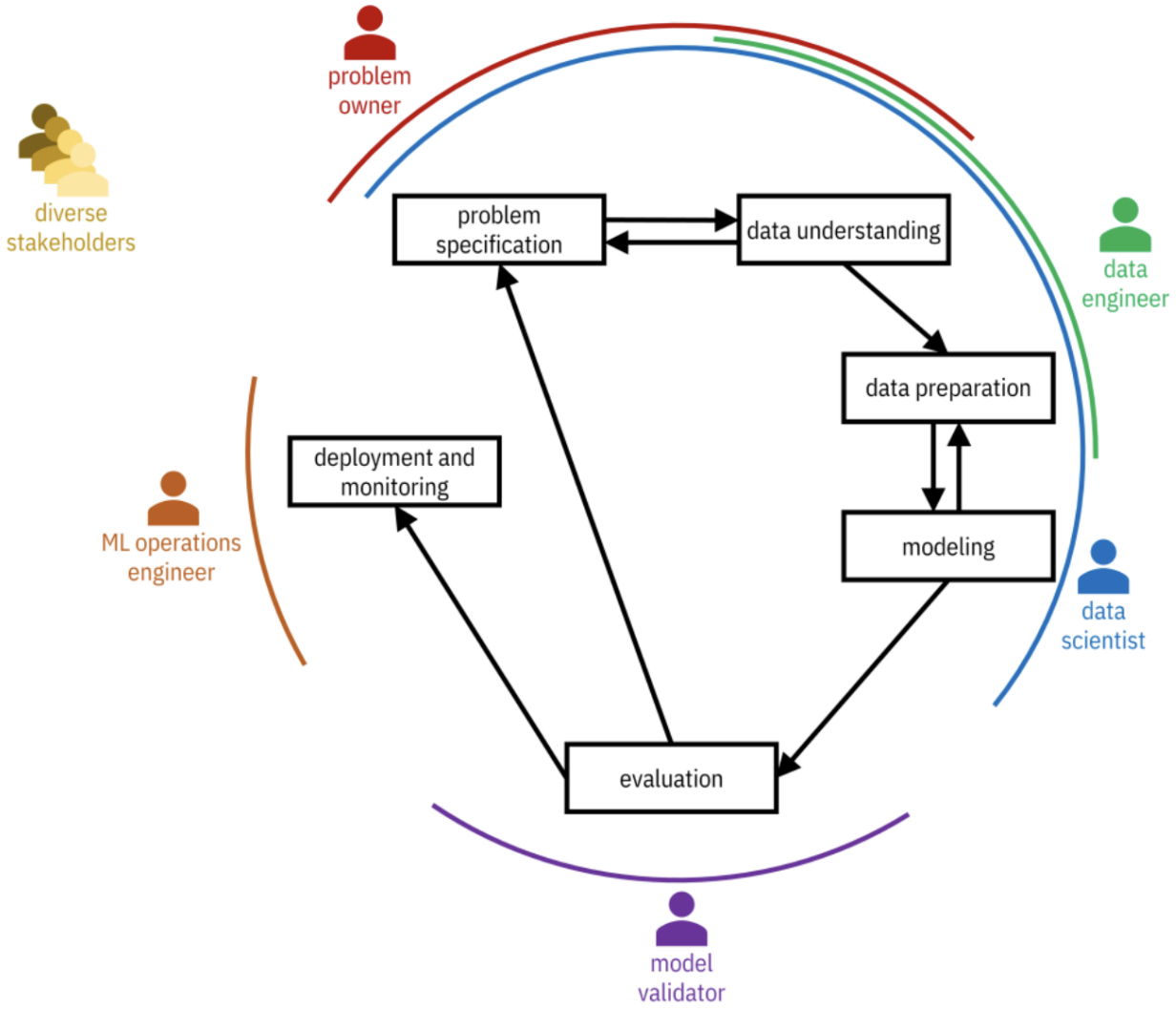
Siklus Hidup Pembelajaran Mesin

General to MLOps

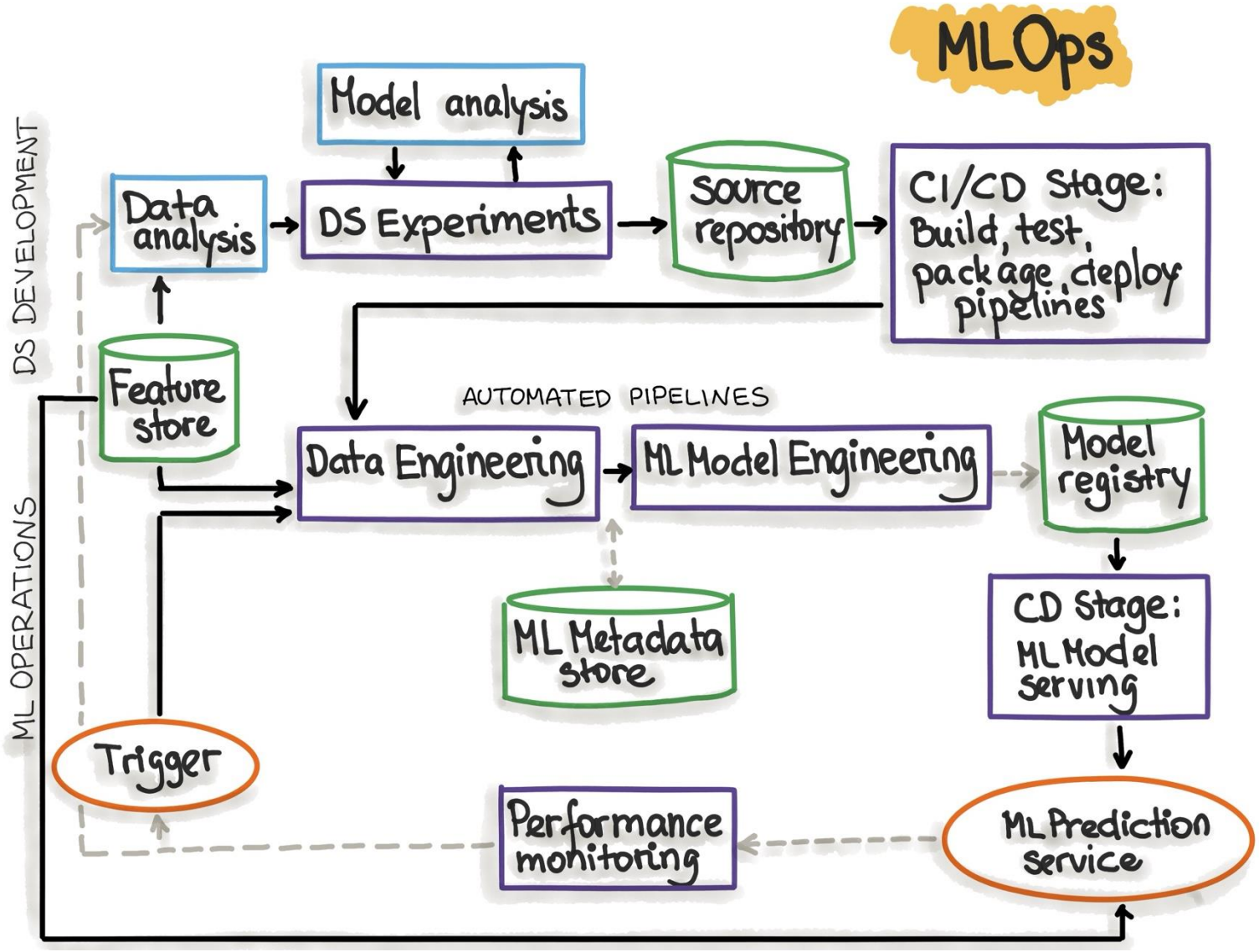
Siklus umum


Steps of building machine learning models
Data access and collection
I
II
Data preparation and exploration
Model monitoring
VI
III
Model build and train
V
Model deployment
IV
Model evaluation

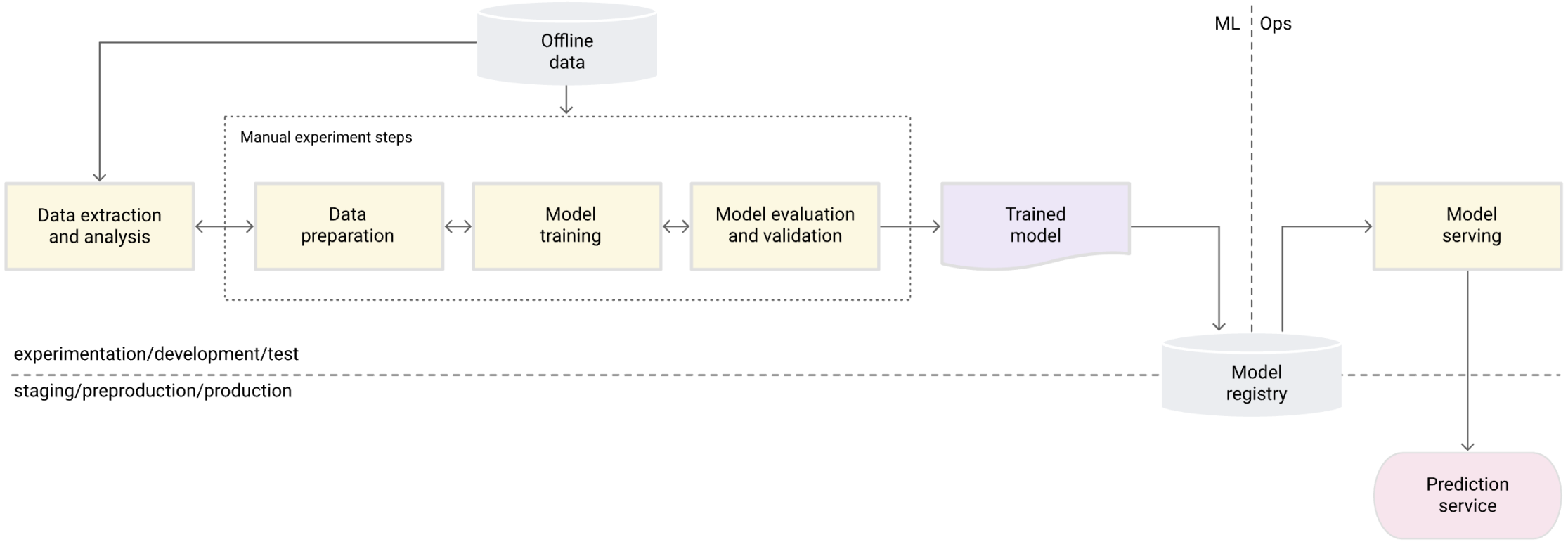
Siklus CRISP-DM



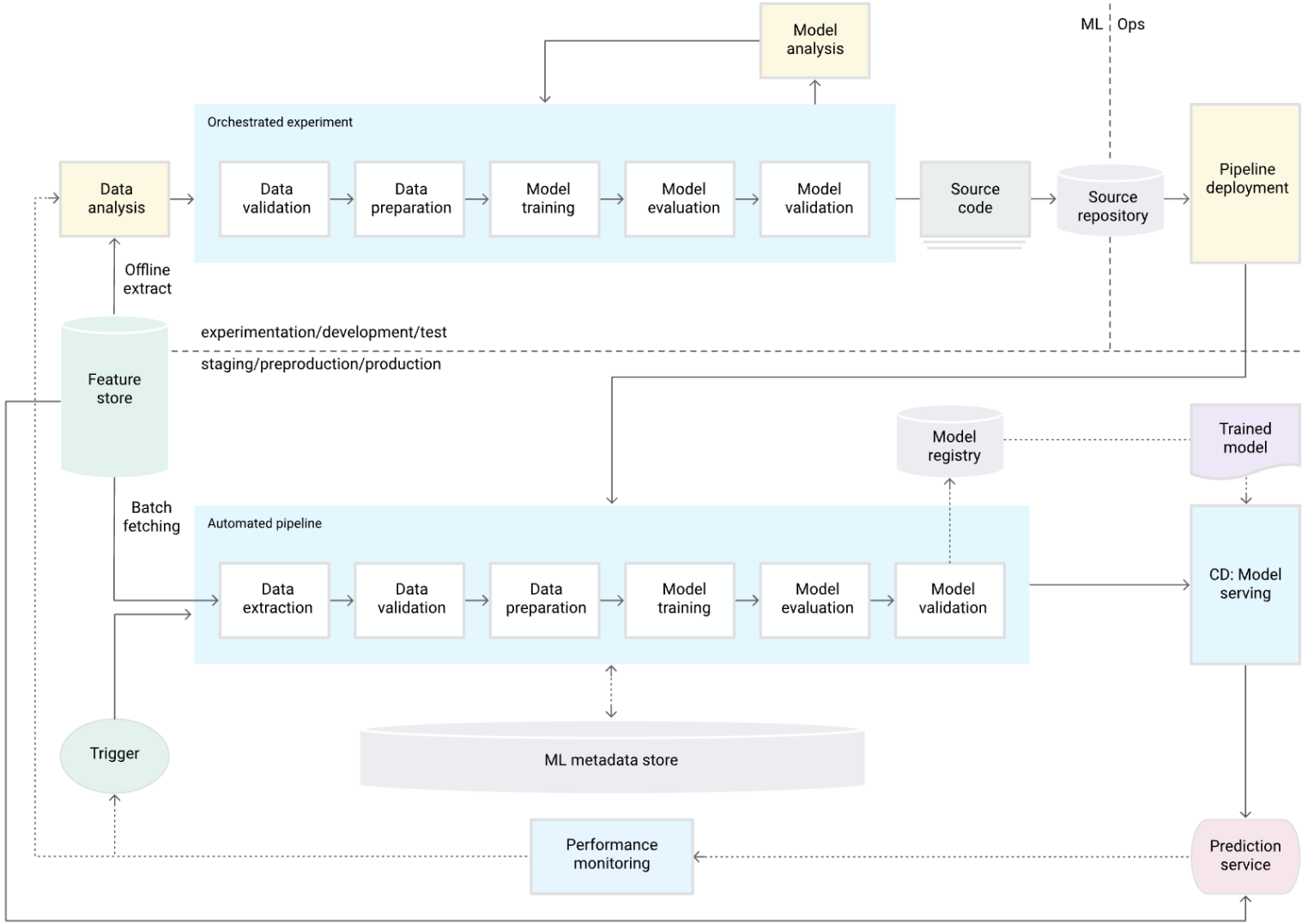
MLOps


MLOps
Model analysis
Data analysis
DS Experiments
Source repository
CI/CD Stage: Build, test, package, deploy pipelines
DS DEVELOPMENT
Feature store
AUTOMATED PIPELINES
Data Engineering
ML Model Engineering
Model registry
ML OPERATIONS
ML Metadata store
CD Stage: ML Model serving
Trigger
Performance monitoring
ML Prediction service

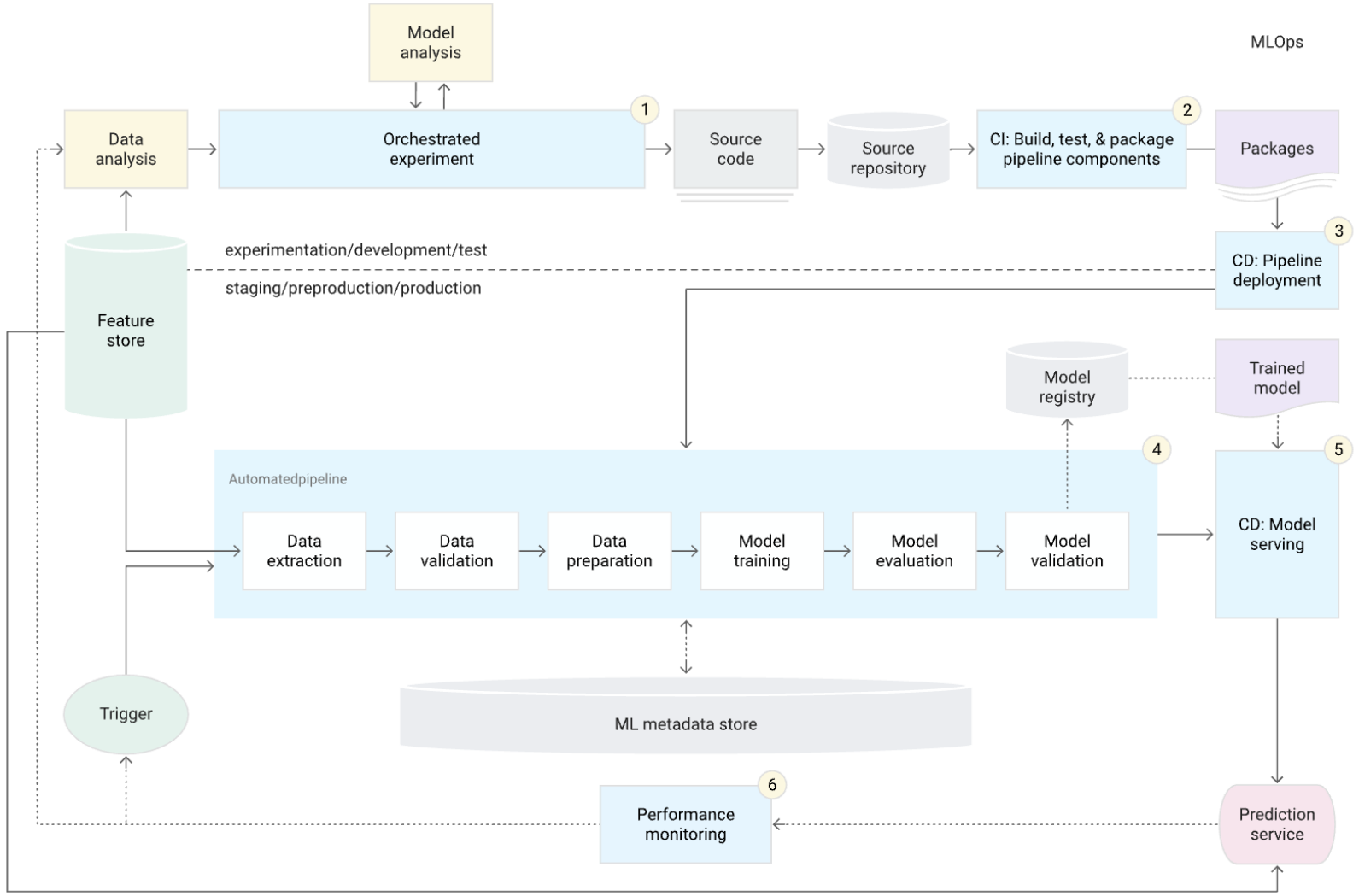
MLOps Level 0 – Manual Process



MLOps Level 1 – ML Pipeline Automation

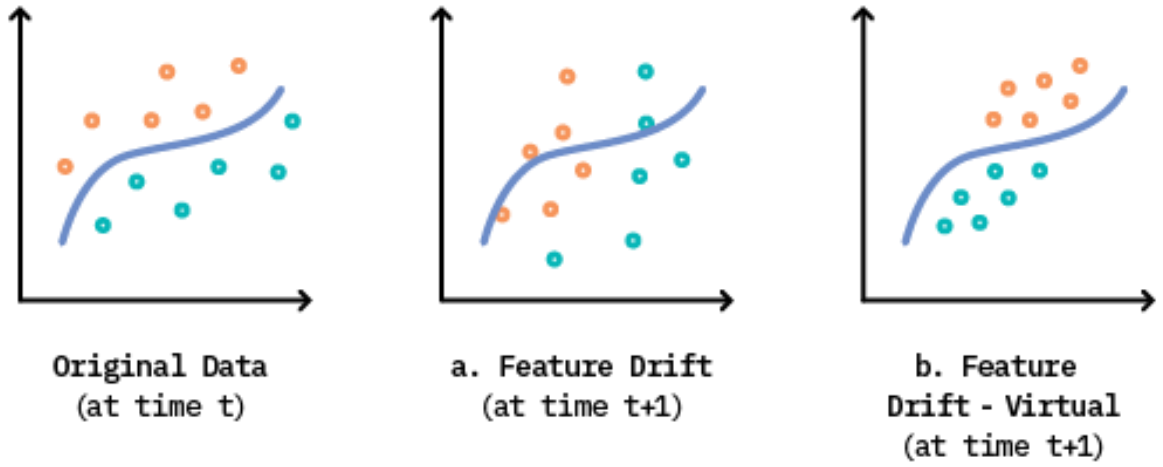


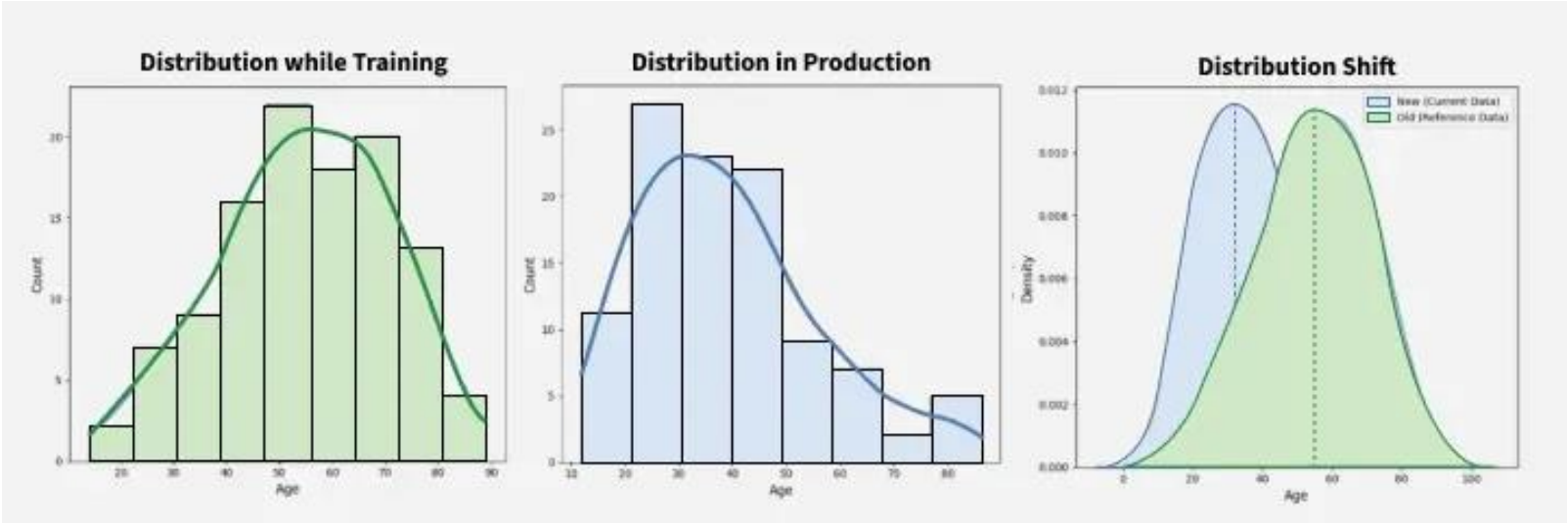
MLOps Level 3 – CI/CD Pipeline Automation



Mengapa perlu dipantau?

Drift in Machine Learning





Etika dan Tantangan

Responsible AI

Etika Pemanfaatan Pembelajaran Mesin

Microsoft AI Principles – Responsible AI

Fairness

AI systems should treat all people fairly

Reliability and Safely

AI systems should perform reliably and safely.

Privacy and Security

AI systems should be secure and respect privacy.

Inclusiveness

AI systems should empower everyone and engage all people, regardless of their backgrounds.

Transparency

AI systems should be understandable.

Accountability

People should be accountable for AI systems.

Tantangan

Pengembangan

- Akurasi data dan bias
- Kurangnya data
- Kurangnya ahli dalam GenAI
- Kurangnya ahli dalam justifikasi keuangan dan bisnis

Privasi dan Keamanan

- Data breaches
- Adversarial attack → Modifikasi model AI dengan memberikan data latih yang tidak tepat
- Model poisoning
- Penyalahgunaaan AI → Deepfake, phising, dst

Isu Lingkungan

- Penggunaan air murni pada proses pembuatan semikonduktor dan pendingin server


תודה
Dankie
Gracias
Спасибо
شكراً
Merci
Takk
Köszönjük
Terima kasih
Grazie
Dziękujemy
Dėkojame
Ďakujeme
Vielen Dank
Paldies
Kiitos
Täname teid
谢谢
Thank You
Tak
感謝您
Obrigado
Teşekkür Ederiz
감사합니다
Σας Ευχαριστούμ
ขอบคุณ
Bedankt
Děkujeme vám
ありがとうございます
Tack

Referensi

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